Storytelling as Inverse Inverse Planning

Chandra, Li, Tenenbaum & Ragan-Kelley (2023) — with Borges, *The Garden of Forking Paths* (1941)

Aldo Costa ▪ Master-Seminar: Theory of Mind, HHU ▪ 21.01.2027

Notes and slides

In one sentence

A storyteller plans for the audience's beliefs, not for the characters' goals: it chooses actions so that the audience's inferences, modelled as Bayesian theory of mind, follow a designed path.

Motivation

Viewers see agents with goals in simple moving shapes (Heider & Simmel, 1944). A rational planner reaches its goals, but an efficient path leaves no doubt: no suspense, no irony. Kukkonen (2014): a story is a "probability design" of the reader's beliefs.

Three levels

Planning goal → actions (act)
Inverse planning actions → $\mathrm{P}(\text{goal} \mid \text{actions})$ (ToM)
Inverse inverse objective → actions, via an audience model

The storyteller plans *through* a model of the audience's theory of mind.

Audience model (Ullman et al., 2009)

Grid world: a robot that can push objects and a weak cheese (its moves succeed 60% of the time), goal tiles pink and green. The robot's reward includes $\rho \cdot r_{\text{cheese}}$, the cheese's reward weighted by ρ: $\rho > 0$ helps, $\rho < 0$ hinders, $\rho = 0$ indifferent. The robot reasons about a cheese that reasons about the robot. An observer infers the hypothesis H (both goals G and ρ) from each action a in state s:

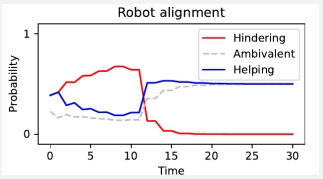
$$\mathrm{P}(H \mid s \to a) \propto \mathrm{P}(s \to a \mid H)\,\mathrm{P}(H)$$

Optimizing a script

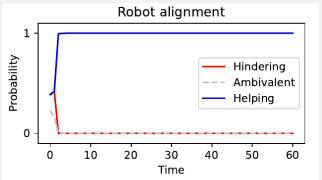
A script σ is an initial state plus a sequence of moves; $\sigma_{1:t}$ is the part seen so far. The setting and the outcome of each move are optimized too. Beam search (keep the best few partial scripts) maximizes, for example,

$$f_{\text{help}}(\sigma) = \sum_t \mathrm{P}(\rho > 0 \mid \sigma_{1:t}).$$

The objective fixes no goals or positions — the optimizer finds a scene that shows helping.



naïve planning: unsure



inverse inverse: helping

Observer's belief about the robot over time (Fig. 1–2).

Limits and critique

- **Paper**: scales badly (state, hypothesis, script spaces); too slow for real-time experiments; emotion is future work.
- **Borges** (my reading): the Chief must already hold the hypothesis "a killing encodes a name"; one lossy report, not a full trajectory.
- **Ethics** (my extension): the same machinery can deceive.

Results

Exp. 1: 98 participants, 120 animations; % who report the intended relation, naïve vs. inverse inverse: helping 6 vs. **73**, hindering 29 vs. **63** (both $p < 0.01$), indifferent 75 vs. 73 (n.s.). Not all scenes allow helping (cheese already near its goal).

Exp. 2 ("inverse inverse physics"): a hopping robot mimes pulling a heavy box. Controls: chance on same-weight pairs, 97% on honest heavy vs. light. The mime is judged heavier than an honest light box (**95.7%**) and than an honest *heavy* box (**68.6%**). 35 participants in the text; Fig. 4b says N = 70.

Storytelling devices as objectives

- **Plot twist**: $\mathrm{P}(\rho > 0)$ in the first half, $\mathrm{P}(\rho < 0)$ in the second.
- **Dramatic irony**: the audience knows the cheese's goal; the robot does not and hinders by mistake.
- **Flashback**: append a known event to σ; find a past that makes it look helpful.
- **Narrative arc**: fortune rises, falls, rises; a KL penalty on belief changes keeps the goals stable; a table can fall "from the sky" that no character expects (deus ex machina).

The Garden of Forking Paths (my reading)

Yu Tsun, a German spy in England, must tell his Chief in Berlin the name of a town. He kills a stranger named Albert; the newspapers report it; the Chief understands and the town of Albert is bombed.

- **Reading 1**: the murder is inverse inverse planning — storyteller Yu Tsun, audience the Chief, observations = newspapers. Terrible in the world, the only act that works for the belief: an extreme mime.
- **Reading 2**: Borges shapes the reader's belief the same way — Yu Tsun calls himself Albert's friend, then shoots (twist).
- **Reading 3**: Ts'ui Pen's novel takes *every* branch; a storyteller keeps *one* script and lets the audience infer the meaning: show, don't tell.

For discussion

1. Write an objective f under which Yu Tsun's murder scores highest. Which term does the work?
2. Is the Chief a Bayesian inverse planner? What prior does he need?
3. Could an LLM be the audience model?

Seminar links: Bayesian ToM 03.12. ▪ LLM inverse planning 17.12. ▪ memo 14.01. ▪ false belief 29.10. ▪ Clever Hans 05.11. ▪ interlocutor awareness 07.01. ▪ epistemic logic 04.02.

Chandra, K., Li, T.-M., Tenenbaum, J. B., & Ragan-Kelley, J. (2023). Storytelling as inverse inverse planning. *Proc. CogSci* 45 (CC BY 4.0); code and videos: https://people.csail.mit.edu/kach/a2i2p/. Borges, J. L. (1941). El jardín de senderos que se bifurcan. Heider, F., & Simmel, M. (1944). An experimental study of apparent behavior. *Am. J. Psychol.* 57. Ullman, T. et al. (2009). Help or hinder. *NeurIPS*. Kukkonen, K. (2014). Bayesian narrative. *Anglia* 132.